

浙江大学



Lab Practice Report for Undergraduate

Name:

College: ZheJiang University

Department: Science

Major: Marine Science

Student NO.:

Instructor: Yanan Di

Date: 2024.11.24

Lab Practice Report of Zhejiang University

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Course Name: Marine biology Type of Practice: Analyze

Name of Practice: Protein assay to determine the protein contents
indifferent marine organisms

Student Name: _____ Major: Marine biology

_____ Student Number:

Team Memeber: Zhiang Liu

Instructor: Yanan Di

Practice Location: 241 Practice Date: 2024.11.24

1. Aims and Requirement of the Practice (Compulsory)

1. Learn the basic procedure to measure protein content in biological samples
2. Understand the basic theory of protein assay
3. Learn the basic procedure to use microplate reader
4. Practice the basic analysis of the measure results

2. Background information of the Practice

The measurement of the amount of a biomaterial substance is usually by means of light absorption, which is based on a comparison of the light absorption of an unknown substance and the light absorption of a known substance.

In this experiment, we used the Bradford protein assay to confirm the concentration of protein in solution, which involves the binding of Coomassie Brilliant Blue G-250 dye to protein. It exists in three forms: cationic, neutral and anionic. The color of the dye changes with the presence of the protein, and we detect this form with a microplate readers at 595 nm.

Microplate readers are the instruments which are used to detect biological, chemical or physical events of samples in microtiter plates. The most common microplate format used is 96-well with a typical reaction volume between 100 μ L and 200 μ L.

3. Equipment and materials

1. Supernatant from tissue samples you have preserved before
2. Microplate reader and 96-well microtiter plate
3. 1.5 mL tube, 50 mL Sterilized test tube,
4. Bradford staining(A) and BSA standard(B)(5mg/mL), PBS solution(C).
5. Pipette and tips with different sizes
6. Marker pens, pencil paper towels, plastic bags, gloves, ice box
7. USB

4. Procedures

1. Sample preparation: Take your preserved supernatants from tissue samples out from freezer. Make sure to keep the supernatant on ice before further application. Mark your sample.

2. Protein extraction(Done in the previous experiment)

3. Protein standard curve preparation(3 standard curves totally):

B is the standard protein of 5mg/mL BSA stock. Use it to make up a series of dilutions to reach the final concentrations are 0.025, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5 mg/mL.

Top up with 200 μ L Bradford in to each well, wait for 3-5 min for the reaction. Using the microplate reader to measure the A_{595nm} in each well with sample. Make the record and then draw the standard curve.

4. Protein and Bradford reagent reaction:

Add 20 μ L of prepare protein sample into the wells of microplate, top-up with 200 μ L Bradford (A) into each well, wait for 3-5min for the reaction. Using the microplate reader to measure the A_{595nm} in each well with sample.

5. Calculation of the protein concentration

Calculate the protein concentration in the supernatant according to the standard curve. Calculate protein by the equation.

Protein concentration (mg/g)=Protein concentration in supernatant(mg/mL)/Wet weight of tissue(g/mL)

5. Question

1. Discuss the basic theory of the protein measurement method. Is there any other method to determine the protein concentration? Give at least two other examples.

Basic theory: In the Bradford method, the color of the dye utilizing three different electrical properties varies with the presence of the protein and is determined with a microplate at 595 nm

other examples:

(1) Lowry Method

The Lowry method is a colorimetric assay based on the reaction of proteins with copper ions. The method first reduces peptide bonds in proteins to divalent copper ions using an alkaline solution, and then adds Folin-phenol reagent to form a blue compound. The absorbance of this compound is proportional to the protein concentration.

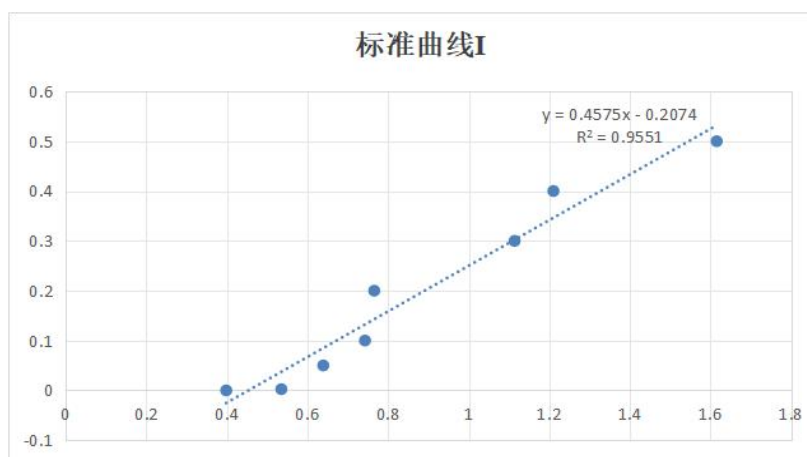
(2) Bicinchoninic Acid Assay

The Bicinchoninic Acid (BCA) assay is a colorimetric method based on the formation of a purple complex between proteins and copper ions. In this method, proteins reduce divalent copper ions to monovalent copper ions under alkaline conditions, which then react with bicinchoninic acid (BCA) to form a purple complex. The absorbance of this complex is proportional to the protein concentration.

2. Make up the standard curve (personal and group). The standard curve from group data should have basic statistics analysis(average, error bars, standard error/standard deviation).

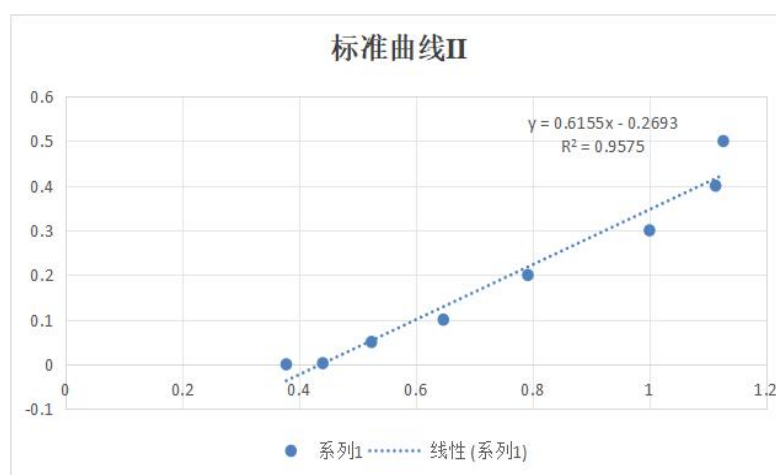
First standard curve (from Zhiang Liu)

1.6156	1.2103	1.1139	0.7655	0.7426	0.639	0.5349	0.3981
0.5	0.4	0.3	0.2	0.1	0.05	0.0025	0



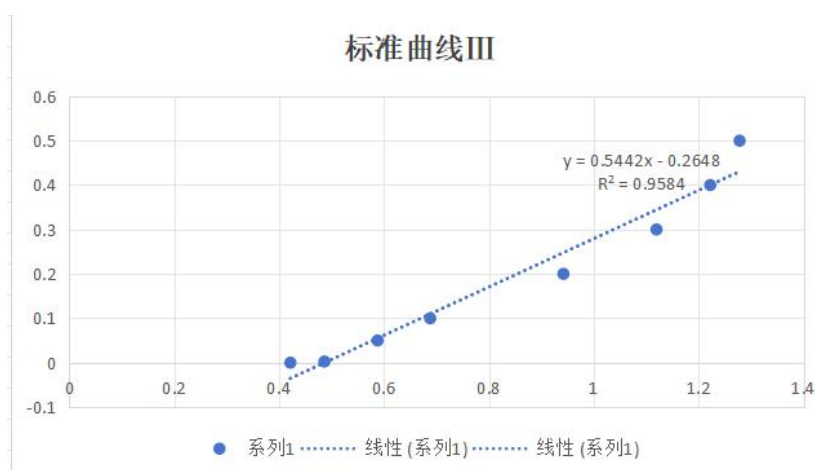
Second standard curve (from Hongyu Ke)

1.127	1.1138	1.0005	0.792	0.6471	0.5239	0.4404	0.3779
0.5	0.4	0.3	0.2	0.1	0.05	0.0025	0



Third standard curve (from Hongyu Ke)

1.2779	1.2219	1.1195	0.9421	0.688	0.5878	0.4864	0.4216
0.5	0.4	0.3	0.2	0.1	0.05	0.0025	0



1. 6156	1. 2103	1. 1139	0. 7655	0. 7426	0. 639	0. 5349	0. 3981	G1
1. 127	1. 1138	1. 0005	0. 792	0. 6471	0. 5239	0. 4404	0. 3779	G2
1. 2779	1. 2219	1. 1195	0. 9421	0. 688	0. 5878	0. 4864	0. 4216	G3
1. 3402	1. 1820	1. 0780	0. 8332	0. 6926	0. 5836	0. 4872	0. 3992	AVERAGE
0. 2502	0. 0593	0. 0671	0. 0952	0. 0479	0. 0577	0. 0473	0. 0219	STDEV

3. Calculate the protein concentrations in each sample and compare your results with your team members, draw figures of your results.

Here's protein static

D1	D2	D3	M1	M2	M3	
0. 4543	0. 5226	0. 443	0. 5522	0. 5761	0. 5619	Shirmp
0. 4259	0. 4521	0. 4385	0. 5229	0. 5712	0. 6375	Crab
0. 5336	0. 5835	0. 5152	0. 8126	0. 8292	0. 8628	Octopus
0. 5457	0. 5352	0. 6016	0. 7118	0. 6064	0. 5805	mussels

D means Digest tissue.

M means Muscle tissue.

RESULT by standard curve II :

D1	D2	D3	M1	M2	M3	
0. 4543	0. 5226	0. 443	0. 5522	0. 5761	0. 5619	Shirmp
0. 4259	0. 4521	0. 4385	0. 5229	0. 5712	0. 6375	Crab
0. 5336	0. 5835	0. 5152	0. 8126	0. 8292	0. 8628	squid
0. 5457	0. 5352	0. 6016	0. 7118	0. 6064	0. 5805	mussels

Concentration

D1	D2	D3	M1	M2	M3	
3. 1159	3. 5795	2. 6369	3. 9025	3. 7385	3. 6654	Shirmp
2. 7407	3. 0966	2. 6101	3. 6954	3. 7067	4. 4487	Crab
3. 3666	4. 0325	3. 5167	5. 2527	5. 5317	5. 8101	squid
3. 8028	3. 7218	3. 5703	4. 9056	3. 9326	3. 9091	mussels

4. Discuss the results and analyse the reasons of such results, pay attention about the average, standard deviation, standard error, etc.

RESULT by standard curve

D1	D2	D3	AVE D	STD D	M1	M2	M3	AVE M	STD M	
0. 0603	0. 1024	0. 0534	0. 0720	0. 0265	0. 1206	0. 1353	0. 1265	0. 1275	0. 0074	Shirmp
0. 0428	0. 0590	0. 0506	0. 0508	0. 0081	0. 1025	0. 1323	0. 1731	0. 1360	0. 0354	Crab
0. 1091	0. 1398	0. 0978	0. 1156	0. 0218	0. 2809	0. 2911	0. 3118	0. 2946	0. 0157	squid
0. 1166	0. 1101	0. 1510	0. 1259	0. 0220	0. 2188	0. 1539	0. 1380	0. 1702	0. 0428	mussels

Concentration

D1	D2	D3	AVE D	STD D	M1	M2	M3	AVE D	STD D	
3. 1159	3. 5795	2. 6369	3. 1108	0. 4713	3. 9025	3. 738481506	3. 665362035	3. 7688	0. 1214	Shirmp
2. 7407	3. 0966	2. 6101	2. 9633	0. 2518	3. 6954	3. 706683971	4. 448709002	3. 8595	0. 4317	Crab
3. 3666	4. 0325	3. 5167	3. 2272	0. 3493	5. 2527	5. 531687792	5. 81010101	4. 7409	0. 2787	squid
3. 8028	3. 7218	3. 5703	3. 6685	0. 1180	4. 9056	3. 932555123	3. 909090909	4. 8903	0. 5687	mussels

From those table, we can see that the protein content of muscle tissue is significantly higher than that of digestive tissue.

Reflected in the specific species, the order of the proteins of the digestive tissue is:

mussel>squid>shrimp>crab.In terms of specific species, the order of muscle tissue proteins is:
squid>mussel>crab>shrimp.

In terms of standard deviation, the standard deviation of crab digested tissue and squid muscle tissue were obviously low, indicating that the error between parallel samples was small during the acquisition of these two samples

6. Discuss if protein contents are altered depending on species and tissues?

There are differences.

From those table, we can see that the protein content of muscle tissue is significantly higher than that of digestive tissue.

Reflected in the specific species, the order of the proteins of the digestive tissue is:
mussel>squid>shrimp>crab.In terms of specific species, the order of muscle tissue proteins is:
squid>mussel>crab>shrimp.

In terms of standard deviation, the standard deviation of crab digested tissue and squid muscle tissue were obviously low, indicating that the error between parallel samples was small during the acquisition of these two samples

注：不同类型的实验课对实验报告可有不同要求，各个学科的实验报告可以根据自己的学科特点做适当的调整，但上述基本内容中的第一、二、三、六条为必须填写的内容。